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CELL BIOLOGY

Autophagolysosomal exocytosis inverts Src kinase onto the cell surface in cancer

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INTRODUCTION: Antibody-targeted therapeutics for cancer (e.g., T cell engagers, antibody-based radioligand therapies, and antibody-drug conjugates) require a cancer-associated protein to be displayed on the cell surface at much higher levels than in healthy tissues. Typically, targets are proteins highly up-regulated by cancer cells for survival (e.g., HER2 and EGFR). However, therapies targeting these proteins are limited by the expression of these cell surface targets in healthy tissues. There is therefore a need to find and understand new classes of tumor-associated cell surface proteins that have greater differences in tumor versus healthy tissue expression. Interest in new targets for cancer therapeutics has led to the discovery of new ways to differentiate tumor cells from healthy cells using their cell surface proteins. These differences include genetic mutations specific to cancer, the activities of enzymes yielding new posttranslational modifications, and aberrant protein localization.

RATIONALE: We developed a system to identify cell surface kinase substrates as well as to define a high-resolution cell surface proteome through photo-proximity labeling proteomics. This method allowed us to both identify protein substrates that get phosphorylated on the cell surface and gain insights into the enzymes mediating phosphorylation.

RESULTS: In this work, we found that Src, the master regulatory kinase typically anchored by its *N*-myristoyl lipid to the inner leaflet of the plasma membrane, was noncanonically translocated to the cell surface through autophagolysosomal exocytosis (ALE). We found additional evidence that other *N*-myristoylated intracellular proteins could be translocated in a similar fashion. Using cell lines with varying levels of cellular Src (cSrc), we observed that the transport of Src to the cell surface was regulated both by autolysosomal flux and total Src protein level. We observed that extracellular Src (eSrc) was present on the surface of cancer cell lines, whole and disaggregated xenografts, and disaggregated patient tumor samples. We propose that the prevalence of eSrc on cancer cells is likely attributable to both cSrc protein up-regulation in cancer and dysregulated autophagic flux, a hallmark of cancer, leading to translocation and inversion of cSrc by ALE. Additionally, we characterized the cell surface phosphotyrosine substrates of eSrc and showed that eSrc activity promoted cell proliferation. Finally, we demonstrated the translational potential of eSrc as a diagnostic and therapeutic cancer antigen using an anti-Src antibody formulated as a bispecific T cell engager (TCE), an antibody-drug conjugate (ADC), or a radioligand therapeutic (RLT).

CONCLUSION: We describe a class of mislocalized proteins, prototypically the proto-oncogene tyrosine kinase Src. These proteins are N-terminal lipid-modified proteins, canonically anchored to the plasma membrane, that undergo topological inversion through a noncanonical secretory pathway that we elucidate in this work. Collectively, these findings uncover a class of proteins by which cancer cells can be therapeutically targeted with antibodies. □

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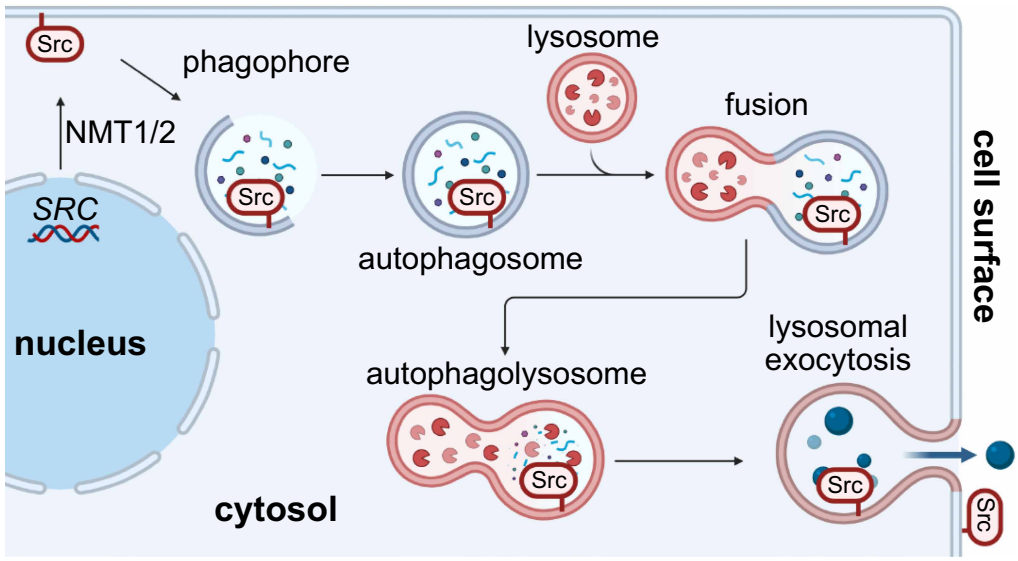

Src is translocated to the cell surface. *SRC* is translated, the protein is lipidated, and the lipid moiety inserts into the inner leaflet of the plasma membrane. Nucleating phagophore membranes originate from plasma membrane components, carrying Src along. After autophagosome formation, lysosomal fusion occurs, yielding autolysosomes, which can be exocytosed resulting in cell surface presentation of Src. [Figure created in part with BioRender.com]

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CELL BIOLOGY

Autophagolysosomal exocytosis inverts Src kinase onto the cell surface in cancer

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Overexpression of the proto-oncogene Src is common to a wide variety of cancers. In this work, we found that Src is noncanonically translocated and inverted onto the cell surface in cancer, both in vitro and in vivo. We identified autophagolysosomal exocytosis (ALE) as a secretory mechanism prominent in cancer cell lines. Src represents the prototypical example of a family of membrane-anchored proteins that are transported by this process. Furthermore, this extracellular membrane–associated Src (eSrc) was found in primary tumors, and anti-Src antibody-based therapies mediated tumor cell killing in cell culture systems and in mouse xenograft models. Thus, intracellular *N*-myristoylated proteins, prototypically Src, can be topologically inverted onto the cell surface in cancer and targeted with antibody therapeutics.

Cancer-targeting antibody therapies typically require either high levels of protein-level up-regulation or the generation of a completely new epitope—e.g., from genetic mutations or aberrant posttranslational modifications (*1*). Several examples have been described where protein mislocalization allows intracellular proteins to translocate to the cell surface by an unknown mechanism or mechanisms (*2*). Such targets defy the paradigm that only classically secreted or membrane-bound targets are viable candidates for immunotherapy.

There is growing evidence that some protein kinases are secreted and have roles in the extracellular space (*3*–*9*). Theoretically, these ectokinases have the potential to hyperphosphorylate extracellular proteins in the tumor microenvironment owing to the high abundance of extracellular adenosine 5′-triphosphate (ATP) in cancer (up to 1 mM) well above typical Michaelis constant (K_m; affinity for substrate) values for kinases; ATP levels in normal tissues are about 10 nM (*10*, *11*). For example, targeting one such ectokinase, CK2, to cells overexpressing HER2 (by fusing CK2 to an anti-HER affibody) led to phosphorylation of several cancer-associated surface proteins, including CD44, Galectin-3, and CD109 (*8*). In this work, we sought to probe more broadly for the presence of ectokinases specifically on the surface of the human breast cancer cell line, HCC1569.

Src remodels the cell surface phosphoproteome

Higher-resolution photo-proximity labeling proteomics (PLP), such as μMap and MultiMap, can be used to trigger short-range reactive biotin-carbene labeling to profile the surface proteome of cells (*12*, *13*). Using PLP, we identified 738 surface proteins, and from this list, we identified 28 protein kinases. Among these was the cancer-associated tyrosine kinase Src as well as other Src family kinases, Fyn, Yes, and Lyn. Src has been found in the media of cancer cell lines through an unknown secretory mechanism(s) (*6*, *7*), but to our knowledge, Src has not been reported on the cell surface.

Given the presence of these tyrosine kinases, we tested whether HCC1569 cells could undergo surface tyrosine phosphorylation. Cells were washed and treated with 1 mM ATP for 30 min at 37°C, conditions typically used for measuring Src activity. After treatment, cells were washed and stained with a selective antiphosphotyrosine (anti-pY) antibody, derived from the commercial clone 4G10 (*14*). We observed increased binding of an anti-pY antibody to the ATP-treated cells; furthermore, most of the labeling was blocked with a small-molecule inhibitor of Src, Src inhibitor 1 (*15*)—a molecule that shows selectivity for Src over other tyrosine kinases (Fig. 1A and fig. S1). These data suggest that most of the surface pY was coming from a Src-like kinase on the HCC1569 cells. Among the Src family kinases detected in our surfaceomics study, Src is reported to have the highest transcriptional level in HCC1569 cells (*16*).

To identify putative Src-like kinase substrates on the surface of HCC1569 cells, we developed a PLP probe using an iridium photocatalyst, enabling short-distance labeling with a diazirine biotin photoprobe (<11 nm) (*17*) coupled to an anti-pY antibody. We validated this approach by comparing it with existing pY proteomic techniques (fig. S2) (*18*). We characterized the surface phosphoproteome of cells treated with or without 1 mM ATP to promote tyrosine phosphorylation and then with ATP and Src inhibitor 1 (Fig. 1, B to F). We identified 631 proteins common among the three independent datasets using label-free quantitation (LFQ). Of these, 53 proteins were specific to the ATP-treated sample set without Src inhibitor 1. There was little difference between the Src inhibitor 1–treated samples and those without ATP, consistent with Src dominating the ATP-induced surface pY phosphorylation that we observed (Fig. 1, D to F).

Among the Src-dependent pY proteins were known Src substrates cortactin (SRC8) (*19*) and LYN (*20*) and the cancer-associated metalloprotease ADAM17 (*21*). We also observed numerous proteins up-regulated by LFQ analyses of the ATP-treated dataset compared with Src inhibitor 1, including cortactin, a known Src substrate, as well as well-known extracellular cancer antigens, such as Erbb2, PTK7, PTPRF, and MUC18/CD146. We hypothesized that some of the targets labeled were intracellular because the proximity labeling from the Ir-pY-Fab generates a diffusible carbene with a labeling radius of 11 nm that is also membrane permeable (*17*).

To validate that Src extracellularly phosphorylates ADAM17, Erbb2, CD44, PTK7, and MUC18, we expressed and purified the ecto-domains for each and treated them either with active Src kinase domain or a loss-of-function mutant, Src(D389A) (*22*). pY was observed for candidates treated with active Src kinase domain but not Src(D389A) by immunoblot using anti-pY (fig. S3, A and B). Upon pY proteomics analysis, we identified one, two, and three pY tryptic peptides for ADAM17, Erbb2, and CD44, respectively, and none when treated with Src(D389A). Each of these pY sites were seen in solvent-exposed positions based on inspection of their respective x-ray structures (fig. S3C). Thus, Src can tyrosine-phosphorylate these proteins, and the PLP method using Ir-pY-Fab is suitable for unbiased ecto-pY protein identification.

We next tested whether extracellular pY phosphorylation by Src would have a functional effect on cells. Treatment of HCC1569 cells with recombinant Src kinase domain, but not the D389A loss-of-function

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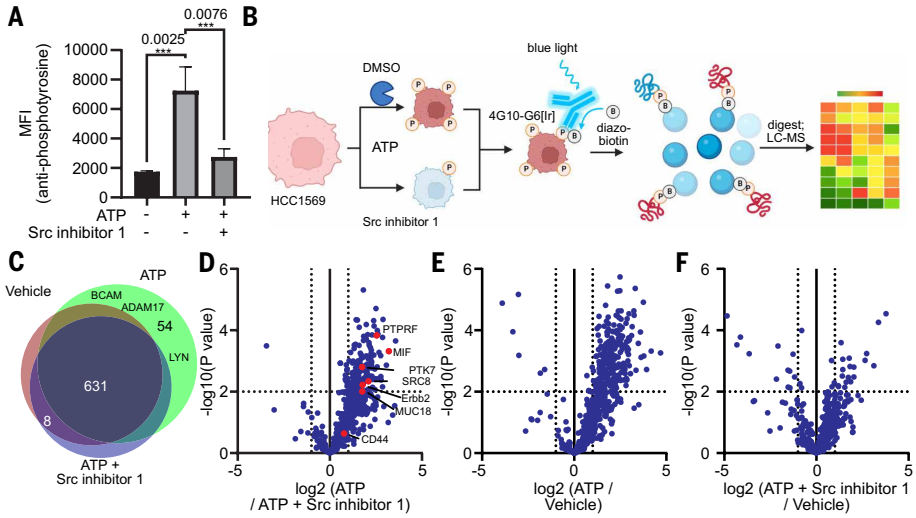

Fig. 1. A Src-like kinase dominates the extracellular tyrosine phosphoproteome in HCC1569 cells. (**A**) HCC1569 breast cancer cells were treated with or without ATP (1 mM) and with or without Src inhibitor 1 (1 μM) to probe eSrc activity. Cells were stained with an anti-pY antibody conjugated with AlexaFluor647 for analysis by flow cytometry. Statistics were calculated by one-way analysis of variance (ANOVA) corrected for multiple comparisons; *P* values are displayed above the relevant comparison. MFI, mean fluorescence intensity. (**B** to **F**) HCC1569 cells were exposed to vehicle, extracellular ATP, or extracellular ATP with Src inhibitor 1. A high-affinity pY-IgG (clone 4G10-G6) (*14*) was coupled to an Ir-based photocatalyst and bound to cells. Irradiation of the Ir-pY-Fab with blue light (460 nm) in the presence of an aryl-diazirine-Biotin photoprobe triggered conversion to a carbene radical, which then labels protein within about 110 Å (*17*). The short half-life of the carbene radical results in short-distance labeling (<110 Å) (*17*), thus strongly biasing the biotinylation of the neighboring pY proteins. Biotinylated proteins were enriched, digested, and analyzed by liquid chromatography–mass spectrometry (LC-MS). (B) Schematic for μMAP-adapted photocatalytic proximity labeling proteomics catalyzing carbene labeling from an anti-pY antibody for identification of eSrc pY substrates. DMSO, dimethyl sulfoxide. (C) Selected known cancer antigens or known Src substrates are annotated. Venn diagram representing proteins with at least two separate peptides identified in at least two of four replicates. Additional comparisons are excluded for brevity. Known Src substrates Lyn, BCAM, and ADAM17 that were found are labeled. (D to F) Protein quantitation based on LFQ of common proteins from (C) and $\log_2$ fold changes were calculated. Only proteins with at least two peptides and shared by both samples are shown. Selected known Src substrate SRC8 and known tumor antigens CD44, MUC18, Erbb2, PTK7, MIF, and PTPRF are highlighted. All error bars represent SDs, and values represent means. All experiments are representative of at least two replicates.

analog, resulted in increased proliferation (fig. S3D). Src activity can enhance the activity of ADAM17 to shed transforming growth factor–α (TGF-α), leading to increased proliferation in a manner independent of the cytoplasmic domain of ADAM17 (*21*, *23*). We hypothesized that ADAM17 could be responsible for the increased proliferation and found that the increase in proliferation induced by active Src was largely abrogated by the ADAM17 inhibitor TAPI-2 (fig. S3E).

Src is found on the cell surface of immortalized cell lines and primary bladder cancer tumor cells

Cellular Src (cSrc) is canonically associated with the inner leaflet of the plasma membrane by an N-terminal myristoyl lipid (*24*–*26*). To establish the localization of an extracellular Src (eSrc), we probed the cell surface (Fig. 2A and fig. S4A) and extracellular media (fig. S4A) of HCC1569 cells with a commercial anti-Src antibody. Src was undetectable in cell media of this cell line, whereas previous studies have detected Src secretion from prostate cancer cell lines (*6*, *7*). However, flow cytometry of live cells showed strong surface expression of Src (fig. S4A), indicating an apparent inversion of cSrc from the inner leaflet of the plasma membrane to the cell surface. Complete knockdown of Src transcription by small interfering RNA (siRNA) was lethal to cells, but partial knockdown was tolerated and reduced cSrc and eSrc expression (fig. S4, B and C).

To determine the generalizability of eSrc cell surface expression, we analyzed surface expression across a variety of cancer-derived cell lines, immortalized models of healthy tissue, healthy patient–derived peripheral blood mononuclear cells (PBMCs), healthy human umbilical vein endothelial cells (HUVECs), and healthy primary bladder epithelial cells (BdECs) (Fig. 2A and fig. S4F). eSrc was expressed by most immortalized cell lines that we tested and was not detectable on healthy donor PBMCs, HUVECs, or BdECs, the latter two harboring very high levels of cSrc (fig. S5). eSrc abundance was not correlated with transcript levels of *SRC* across cell lines [coefficient of determination (R^2) < 0.1], which is not surprising because protein and RNA levels often do not correlate highly (fig. S4D).

We probed the total Src protein levels by immunoblot from select cell lines (as well as HUVECs, PBMCs, and BdECs) and found that eSrc expression was highly correlated (R^2 = 0.92) with total Src protein expression among immortalized and cancerous cell lines (Fig. 2B, fig. S4F, and fig. S5). This trend did not extend to healthy HUVECs and BdECs. This result is notable because Src total protein expression is highly up-regulated in a wide variety cancers (fig. S6) (*27*–*29*), which implies that eSrc may represent a broadly applicable tumor-associated antigen.

In isogenic cell lines, eSrc expression correlated with cell surface tyrosine phosphorylation activity that was inhibited by the Src inhibitor 1, indicating that eSrc is catalytically active (Fig. 2, C and D). Thus, eSrc is expressed and active on the surface of a variety of cancer cell lines. Notably, we did not observe changes in eSrc expression upon exposure to extracellular ATP in the timescale used for the proteomic studies (fig. S4E)

To interrogate the clinical relevance of the inversion of cSrc to eSrc, we obtained peripheral fragments of primary bladder tumors from transurethral removal of bladder tumors (TURBT) as part of clinical treatment. Bladder cancer was chosen owing to high Src protein levels reported in public datasets (*30*) and the up-regulation of *SRC* in transformed cells versus paired normal tissue (GSE223971) (fig. S6) (*30*–*32*). We mechanically dissociated tumor cells and lymphocytes from tissue fragments and analyzed live cells by flow cytometry (Fig. 2A and fig. S7A). All 10 primary tumors that we tested expressed eSrc on live tumor cells. Patient samples robustly expressed Src by immunohistochemistry (IHC) (fig. S7B). Furthermore, the expression of eSrc in patient samples had a comparable proportionality to eSrc levels in bladder cancer cell lines (Fig. 2A), and eSrc modestly correlated with cSrc among the cell lines and patient samples tested but not primary healthy bladder epithelial cells (fig. S7C).

Src is translocated to the cell surface by autophagolysosomal exocytosis

We hypothesized that the *N*-myristoyl lipid on the N terminus of Src (*24*–*26*) should be necessary for eSrc to be displayed on the outer cell membrane. To test this hypothesis, we cultured high-eSrc-expressing HCC1569 cells in the presence of a nontoxic inhibitor of the two human *N*-myristoyl transferases (IMP-1088) (*33*). IMP-1088 did not alter total cSrc protein levels (fig. S8, A and B) but substantially reduced eSrc (Fig. 3A). This was also observed in bladder cancer cell lines (fig. S8C). To confirm that these observations were due to *N*-myristoylation

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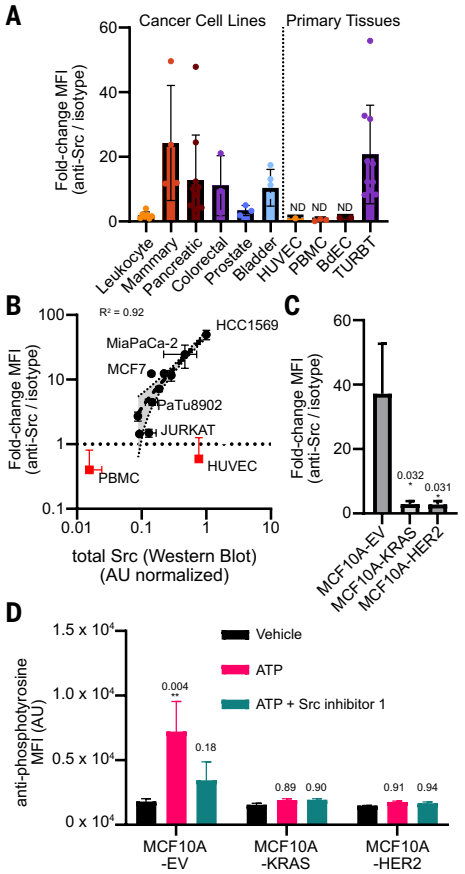

Fig. 2. Src is presented and catalytically active on the surface of cancer cells. (**A**) Cells from various tissues and pathologies were stained with a commercial flow cytometry–validated anti-Src antibody and analyzed by flow cytometry. Each point represents a single cancer cell line (measured in technical replicate), donor (for normal tissues), or patient (for primary samples). ND, not detected above isotype. (**B**) Normalized protein amounts of whole-cell lysates were separated by SDS–polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to nitrocellulose, probed with anti-Src antibody by immunoblot, and detected using a near-infrared dye–conjugated antibody. Normalized total Src fluorescence by immunoblot plotted against the MFI of eSrc detected by flow cytometry. Linear regression analysis indicates a high degree of correlation between eSrc and total Src among immortalized cell lines ($R^2 = 0.92$) but not two normal tissue controls (PBMC or HUVEC). (**C**) MCF10A isogenic cell lines bearing empty vector, oncogenic KRAS, or oncogenic HER2 genes were stained with anti-Src, and eSrc abundance was measured by flow cytometry. (**D**) MCF10A isogenic cell lines with varying levels of eSrc were incubated with or without ATP (1 mM) and with or without Src inhibitor 1 (1 μM). Cells were then stained with an anti-pY antibody to quantify extracellular surface phosphotyrosine abundance by flow cytometry. Statistical comparisons are shown relative to respective control conditions by one-way ANOVA. All error bars represent SDs, and values represent means. All experiments are representative of at least two replicates. AU, arbitrary units.

of Src and not changes from global blockade of *N*-myristoylation, we also transiently transfected low-eSrc-expressing PaTu8902 cells with either C-terminally FLAG-tagged Src or Src(G2A), which cannot be *N*-myristoylated (*26*). Compared with mock-transfected cells, we observed both FLAG staining and increased eSrc staining of wild-type Src-transfected cells but not cells transfected with Src(G2A) (Fig. 3B and fig. S9, A and B). By contrast, transient overexpression of the same construct in primary HUVECs did not result in visible surface staining, congruent with the proposed cancer-specificity of protein inversion (fig. S9, C to E). Thus, eSrc is anchored to the cell surface by its *N*-myristoyl lipid.

Prior work has observed that the lysosomal membrane proteins, LAMP1 and LAMP2, were two of the most enriched surface proteins in empty vector compared with *KRAS*- or *HER2*-transformed MCF10A cells (fig. S10A) (*34*). LAMP1 and LAMP2 can be localized to the cell membrane by lysosomal exocytosis (*35*). Accordingly, transcript levels of transcription factor EB (TFEB), a known transcriptional regulator of lysosomal exocytosis (*36*), are likewise up-regulated (fig. S10B). Previously published surface proteomics studies have identified up-regulation of lysosomal membrane proteins and Src when cultured in hypoxic environments (fig. S10, E and F) (*37*), which is known to stimulate lysosomal exocytosis (*38*). Autophagosomes nucleate inside the cytosol from curved membranes called phagophores to encapsulate cytoplasmic contents and traffic to the lysosome for degradation (*39*–*42*). We hypothesized that if cSrc is anchored to phagophore membranes during nucleation, it would be anchored to the inner (and outer) leaflet of the autophagosome. Upon autophagolysosomal exocytosis (ALE), Src on the inner leaflet of the autophagosome would be exposed on the outer leaflet of the plasma membrane, whereas Src on the outer leaflet of the autophagosome is recycled back to the inner leaflet of the plasma membrane. Both autophagy and lysosomal exocytosis are highly up-regulated in cancer (*43*–*45*), which suggests that eSrc and other *N*-myristoylated proteins may be tumor-associated antigens.

To test this hypothesis, we cultured HCC1569 cells in the presence of sublethal concentrations of Vacuolin-1 or Bafilomycin A1, both of which have numerous effects on lysosomal maturation, including lysosomal exocytosis. Both compounds reduced eSrc, whereas inhibitors of exosome biogenesis, endosomal exocytosis, or blebbing did not increase eSrc (Fig. 3C and fig. S11, A to D). Furthermore, an inhibitor of autophagosome formation, ATG7-IN-2 (*46*), abolished the appearance of eSrc in HCC1569 cells without affecting total cSrc protein levels (Fig. 3C and fig. S11, A to D). Collectively, these data implicate the importance of an autophagolysosomal pathway, which we hypothesized to be ALE, in the secretion of eSrc.

On the basis of these data, we hypothesized that we could also pharmacologically up-regulate eSrc by inducing ALE or increasing total Src expression. To facilitate gain-of-signal studies, we used the low-eSrc-expressing PaTu8902 cells and cultured them in the presence of rapamycin, to induce autophagy; subtoxic concentrations of trametinib, which induces both autophagy and lysosomal exocytosis; or dasatinib, low doses of which elicit an increase total Src expression (*47*) (Fig. 3D and fig. S11E). Trametinib and dasatinib, but not rapamycin, increased the expression of eSrc. This difference could be because ALE is promoted by trametinib, whereas autophagy alone followed by lysosomal degradation is promoted by rapamycin. Indeed, in isogenic MCF10A cells, eSrc levels correlated with lysosomal exocytosis, as evidenced by LAMP1 expression being higher in empty vector compared with KRAS transformants (*34*), as opposed to autophagic flux, which was up-regulated in KRAS-transformed cells (fig. S10, C and D). Hypoxia is also known to up-regulate lysosomal exocytosis, which also increased eSrc (fig. S10, E and F). Thus, during autophagosome biogenesis, cSrc from the plasma membrane (or newly *N*-myristoylated from the cytosol) ends up on both sides of autophagosomes before they close; these are trafficked to and fuse with lysosomes, resulting in Src on the inner leaflet of autolysosomes. eSrc is then presented on the plasma membrane along with other lysosomal markers by lysosomal exocytosis.

We probed for the presence of Src on autophagosomes using confocal microscopy by analyzing the cellular distribution of both Src and LC3A/B in chloroquine-treated cells (Fig. 3, E and F, and fig. S12). Low doses of chloroquine block autophagosome-lysosome fusion, which facilitates imaging by causing autophagosome swelling without inducing apoptosis and restricts colocalization of lysosomal Src and LC3

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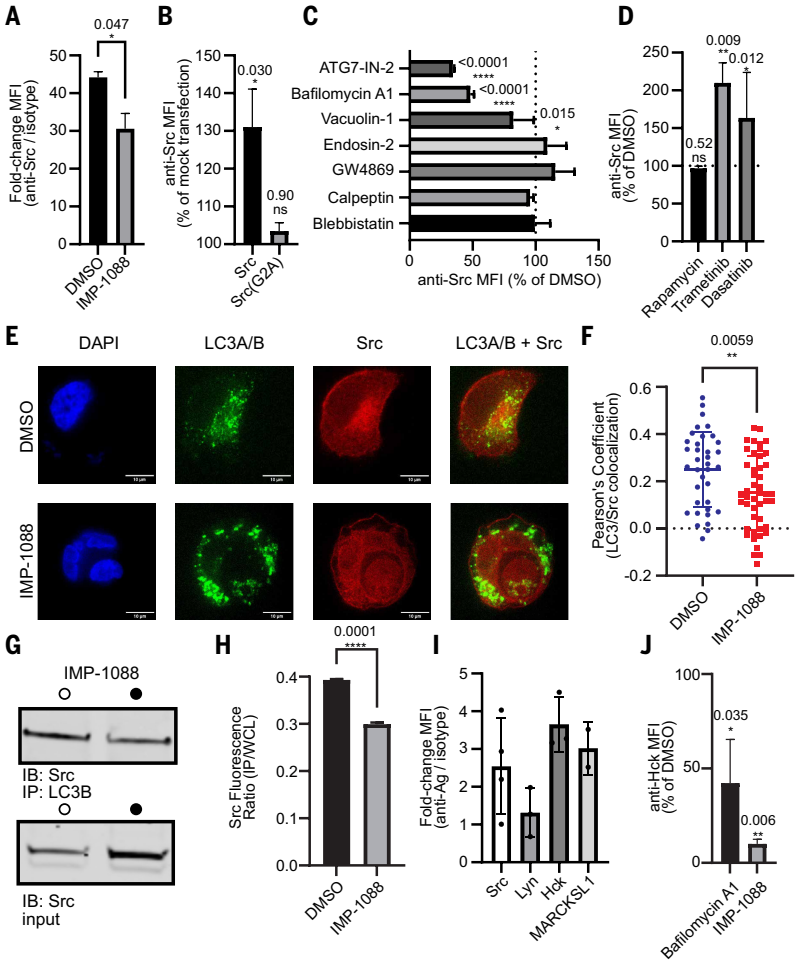

Fig. 3. eSrc is associated with autophagosomes and displayed on the cell surface by ALE. (**A**) HCC1569 cells were treated with IMP-1088 (100 nM) for 72 hours before analysis by flow cytometry. (**B**) PaTu8902 cells were transfected with Src-FLAG or Src(G2A)-FLAG. Live cells were analyzed by flow cytometry for Src and FLAG surface expression 48 hours posttransfection. (**C**) HCC1569 cells were treated for 72 hours before analysis of eSrc surface expression by flow cytometry. (**D**) PaTu8902 cells were treated for 48 hours before analysis by flow cytometry. (**E**) HCC1569 were cultured in the presence or absence of IMP-1088 (100 nM) and transferred to treated chambered glass slides in the presence of chloroquine (50 μM). Cells were fixed, stained, and imaged by confocal microscopy. (**F**) Quantitation of colocalization from multiple fields of view, as described in (E), from two independent experiments. Each data point represents a single cell. Colocalization analysis was performed in ImageJ using threshold colocalization to afford Pearson's correlation coefficient. (**G**) HCC1569 cells were cultured in the presence or absence of IMP-1088 (100 nM) for 72 hours before treatment with Bafilomycin A1 (10 nM) for 2 hours. Cells were harvested and lysed with a Dounce homogenizer, and intact autophagosomes were immunopurified using an antibody that recognizes autophagosome marker LC3B bound to Protein A/G magnetic beads. Autophagosomes were dissolved with 1% NP-40 in tris-buffered saline (TBS), and the contents were analyzed by immunoblot to show Src association. (**H**) Quantitation of two replicates as described for (G). (**I**) Lymphoma cell lines were probed with commercial flow cytometry antibodies and analyzed by flow cytometry. Each point represents the mean of technical replicates of an individual cell line. (**J**) Raji B cells were cultured in the presence of Bafilomycin A1 (10 nM), IMP-1088 (100 nM), or an equal concentration of DMSO for 48 hours before analysis by flow cytometry. For (A), (B), (C), (D), (H), (I), and (J), statistics were calculated by a paired one-way ANOVA on raw data. For (E), statistics were calculated by a mixed model effect. For (H), statistics were calculated by a two-way *t* test. All error bars represent SDs, and values represent means. All experiments are representative of at least two replicates. IB, immunoblot; IP, immunoprecipitation; WCL, whole-cell lysate; DAPI, 4′,6-diamidino-2-phenylindole.

after autophagosome-lysosome fusion. We hypothesized that a small, transient subpopulation of Src could associate with autophagosomes en route to the cell surface and that this colocalization would be sensitive to IMP-1088. As expected, Src appeared across multiple cellular compartments, whereas LC3 was solely observed in swollen puncta. Colocalization analysis between LC3 and Src in chloroquine-treated cells yielded a Pearson's correlation coefficient of $r = 0.25$, indicating a modest degree of colocalization. Also, inhibition of *N*-myristoylation with IMP-1088 resulted in a lower degree of colocalization ($r = 0.15$, $P = 0.0059$, Cohen's $d = 0.66$). Thus, Src colocalization with autophagosomes is dependent on *N*-myristoylation.

To confirm that Src associates with autophagosomes, we used an antibody against the autophagosomal marker LC3B (*40*, *42*, *48*) to isolate intact autophagosomes from HCC1569 homogenates and analyze the contents by immunoblot (Fig. 3, G and H, and fig. S13). We detected Src signal in isolated autophagosomes, which was partially abrogated when cells were treated with IMP-1088 before vesicle isolation ($P = 0.001$). This supports the idea that Src is associated with autophagosomes and that the *N*-myristoyl group can mediate this association.

Src is the prototypical member of a class of noncanonically translocated *N*-myristoylated proteins

On the basis of our proposed mechanism, we hypothesized that other *N*-myristoylated proteins could be noncanonically displayed outside by the same mechanism. In our surfaceomics data, we identified 34 known myristoylated proteins, most of which have not been annotated as extracellular. We independently validated that three other inner leaflet *N*-myristoylated proteins (Lyn, MARCKS, and OAS2) were on the cell surface using specific antibodies to each and flow cytometry of live cells (fig. S14, A to D).

We investigated the generality of these putative inverted proteins by probing the surfaces of leukemia and lymphoma cell lines (Fig. 3I and fig. S14E). The cell lines tested have a low-level expression of eSrc but robust expression of the Src family kinase Hck as well as the unrelated *N*-myristoylated protein MARCKSL1. We identified the inverted form of these proteins (i.e., eHck and eMARCKSL1) on multiple leukemia and lymphoma cancer cell lines but not on healthy donor PBMCs. Furthermore, we demonstrated that eHck expression on the Burkitt's lymphoma cell line Raji was pharmacologically inhibited by IMP-1088 and Bafilomycin A1 (Fig. 3J). We thus propose that eSrc is the prototypical member of a class of inverted proteins tethered to the cell surface by a lipid anchor and translocated by ALE.

A synthetic antibody enables detection and targeting of tumor-associated eSrc in vivo

The combination of cancer-associated Src expression and increased lysosomal exocytosis that drives eSrc translocation suggests that eSrc may be a viable tumor-specific antigen. To begin to assess this, we used our Fab-phage libraries to isolate a recombinant antibody to the human Src kinase domain (Ab1) (*49*–*51*). Ab1 had a single-digit nanomolar dissociation constant (K_d) (2 ± 0.1 nM) for binding either a human or murine Src kinase domain, which share >98% sequence identity, as well as to full-length Src containing the SH2-SH3-kinase domains (Fig. 4A). Ab1 bound eSrc-expressing cells, and binding was reduced when *SRC* expression was partially knocked down to a nonlethal level by siRNA (Fig. 4, B and C, and fig. S15A). Furthermore, we were able to use Ab1 to detect eSrc

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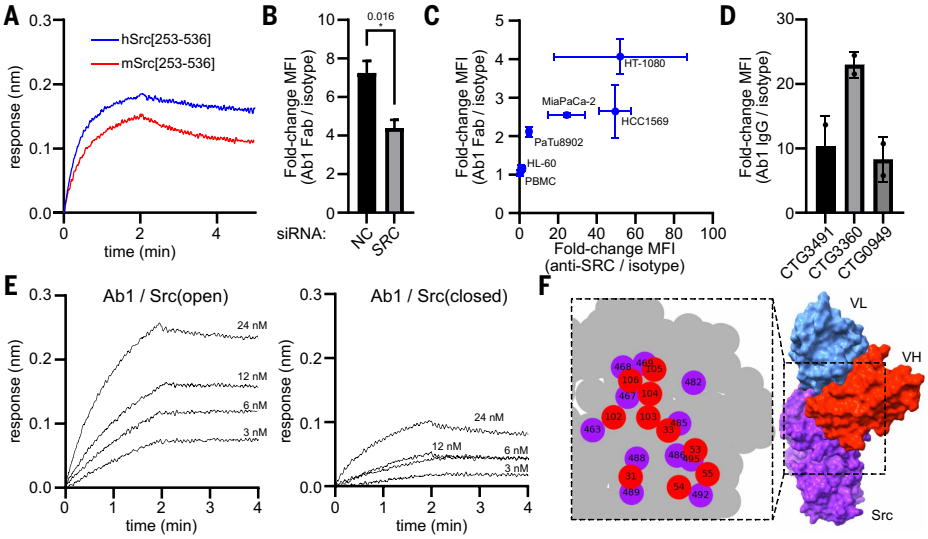

Fig. 4. A selected recombinant anti-Src antibody (Ab1) shows high specificity in vitro and ex vivo. (**A**) Single-point biolayer interferometry curves for Ab1 IgG binding to hSrc and mSrc. Kinetics were calculated in Prism 12. $R^2 > 0.98$ for each curve, affording dissociation constants of 0.9 nM and 3.5 nM for human and murine Src kinase domains, respectively. (**B** and **C**) Ab1 was assayed for binding to cells using its fragment of antigen binding domain (Fab) with a FLAG tag on the C terminus of the light chain and detected with a fluorescent anti-FLAG antibody. (B) HCC1569 cells were transfected with a commercially validated negative control (NC, ucguuaaucggcuauaauacgc) or *SRC* targeting siRNA. Statistics were calculated using Student's *t* test. (C) Binding of Ab1 versus a commercial anti-Src antibody across multiple cell lines. (**D**) Ab1 in a hIgG1 format was used to demonstrate eSrc expression on patient-derived xenografts of colorectal cancer grown in NSG mice. Each dot represents an individual tumor. (**E**) Binding kinetics for the Ab1 IgG1 were determined by biolayer interferometry (BLI) on conformationally locked open or closed mutants of Src. Multipoint BLI traces are shown for Ab1 IgG to either a locked-open mutant (left) or locked-closed mutant (right). Multipoint curve fittings using a bivalent analyte model calculate $K_d = 2.0$ nM for Ab1/Src (open) and $K_d = 647$ nM for Ab1/Src (closed), demonstrating a strong bias toward the open active conformer. (**F**) Cross-linking mass spectrometry predicts a C-terminal epitope near the autophosphorylation loop and C-terminal SH2 ligand. Ab1 was cross-linked with DSSO to recombinant Src. Proteins were digested with trypsin, and cross-links were identified by mass spectrometry. Cross-links were used to predict a complex structure of the Ab1 variable domain (Fv) with Src. The model was then docked in Rosetta. Contact plot (left) shows residues on Src (purple) and Ab1 (VH in red, VL in blue) in contact. Docked structure (right) depicts the bound complex model. All error bars represent SDs. All experiments are representative of at least two replicates.

on fine needle aspirates of resected patient-derived colorectal cancer xenograft tumors (Fig. 4D).

We sought to determine whether eSrc is active on the surface of cells and its copy number per cell. We also generated known conformationally locked mutants of Src in its open or active versus closed forms (*15*, *52*). Notably, we found Ab1 binds preferentially to the open or active conformer of Src and >100-fold weaker to the closed form (K_d = 647 nM) (Fig. 4E). We next determined the copy number of eSrc on HCC1569 cells using a fluorescence quantitation assay with both Ab1 and a commercial anti-Src antibody that binds both the open and closed forms of Src via the N-terminal domain (fig. S15, C and D). Ab1 and the pan-conformation specific Src antibody detected ~25,000 copies of eSrc on the surface of HCC1569 cells, indicating that eSrc was either entirely in the open or active conformation or in a saturable equilibrium. Proteomics-based measurements (*53*) estimate that in healthy murine tissue, Src is expressed around 50 to 500 copies per cell, whereas cancerous cells contain up to 500,000 copies. Thus, even though eSrc is present in cancer cells at levels 20-fold below the level of cSrc, it markedly exceeds cSrc in healthy cells by up to two orders of magnitude, providing a large potential therapeutic index.

Intrigued by the selective binding of Ab1 to the active form of Src, we characterized the epitope on Src for binding Ab1 using cross-linking mass spectrometry (Fig. 4F and fig. S16A) (*13*, *54*). We generated a model of Src in complex with Ab1 based on constraints of intermolecular cross-links, the top 10 of which converged on a single epitope. We validated the predicted epitope by alanine-scanning mutagenesis of two residues on Src and five key contact residues on Ab1, all seven of which reduced affinity measured by biolayer interferometry by three- to >100-fold (fig. S16, B to E). The epitope is near the C-terminal tail, one of the structural determinants controlling the open and closed conformers (*52*), which could thus explain the conformational preference of Ab1.

Given that Ab1 cross-reacts with human and mouse Src, we tested whether Ab1 could bind to eSrc on tumors in mice. We conjugated Ab1 to a heavy metal chelator, deferoxamine (DFO), and then complexed a positron emission tomography (PET) active isotope ^{89}Zr(IV) to the DFO. ^{89}Zr-Ab1 was injected into tumor-bearing Balb/c mice containing aggressive subcutaneous syngeneic EMT-6 murine mammary epithelial cell tumors. We observed high tumor localization of the PET tracer as well as expected uptake into the liver and kidney as the circulating antibody was metabolized. Tumor localization was both intense and sustained, with detectable ^{89}Zr signal past 168 hours and high tumor localization compared with other organs and tissues (Fig. 5A). Furthermore, coadministration of a 100-fold excess of unconjugated Ab1 potently blocked tumor-specific uptake of ^{89}Zr-Ab1 but did not change the ^{89}Zr signal in any other metabolizing tissues (Fig. 5, B and C, and fig. S17A). We observed similar results using subcutaneous HT-1080 cell line tumor xenografts (Fig. 5, D to F, and fig. S17, B and C) and uptake in a bladder cancer patient–derived xenograft (fig. S17, D and E). In all three tumor models, we observed highly specific tumor uptake (<1% false discovery rate), with nonspecific uptake in tissues involved in metabolizing the PET tracer. Thus, eSrc is expressed on tumors and has undetectable endogenous expression on healthy cells in mice.

eSrc is a target for antibody-based therapies

After demonstrating that eSrc is tumor associated in vivo, we hypothesized that Ab1 could be used for antibody-based cancer therapies. To investigate this possibility, we engineered Ab1 in multiple therapeutic formats to evaluate both the accessibility and efficacy of eSrc as a target. Based on the robust uptake of our PET tracers, we hypothesized that a conjugate to a radioactive isotope, such as the alpha-emitting ^{225}Ac, coupled to the antibody through the chelator—e.g., macropa (*55*)—may enable potent killing as a radioligand therapy (RLT) because only a few alpha-particle emissions are required per cell for cytotoxicity (*56*), We tested the efficacy of an ^{225}Ac-Ab1 conjugate against eSrc-expressing HT-1080 tumor cells in a xenograft mouse model (Fig. 5G and fig. S18). Even at 0.6 μCi of ^{225}Ac, we observed either tumor regression (2/16) or a substantial reduction in tumor growth rate (14/16) compared with saline control. We did not observe any acute toxicity or weight loss upon initial dosing; however, liver toxicity was observed upon subsequent doses, consistent with the known general radiation tolerance of mice (fig. S18C).

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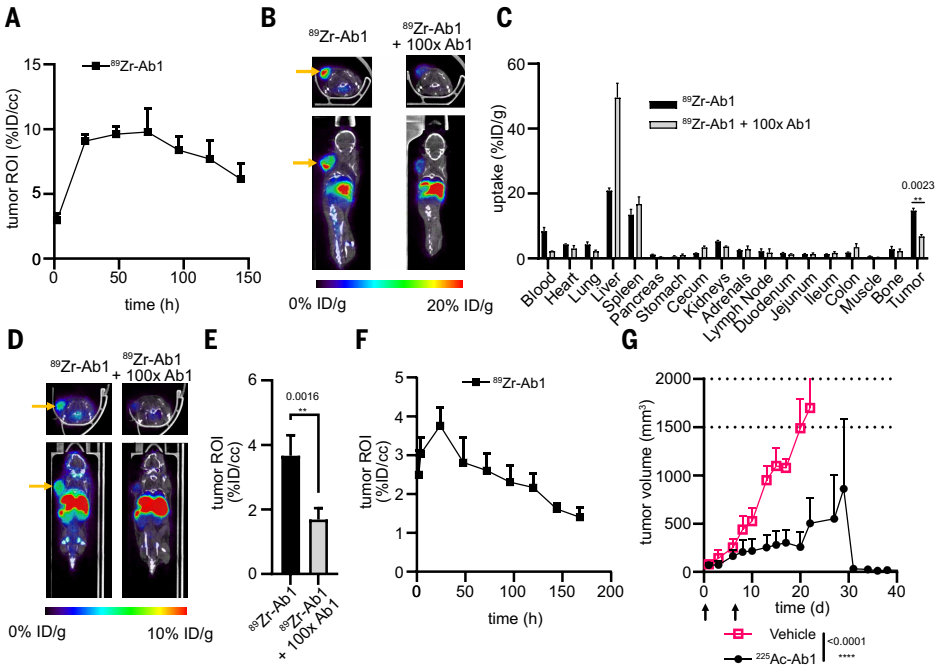

Fig. 5. A recombinant anti-Src antibody (Ab1) localizes to tumors in vivo. (**A**) Ab1 was conjugated to DFO and was then loaded with radioactive ^{89}Zr. Balb/c mice bearing subcutaneous syngeneic EMT-6 murine mammary epithelial tumors were injected with 150 μCi of radioligand-antibody conjugate. Mice were imaged periodically over the course of 6 days. ROI, region of interest. (**B**) As in (A), with or without 100-fold excess unlabeled Ab1 competitor. Representative images of EMT-6 tumor–bearing mice with (right) or without (left) blocking Ab1 24 hours after ^{89}Zr-Ab1 injection. (**C**) Ex vivo biodistribution data of ^{89}Zr from EMT-6 tumor–bearing mice 24 hours after administration of ^{89}Zr-Ab1 with or without 100-fold excess Ab1. Data are averaged values from four animals. (**D**) As in (B), with HT-1080 xenograft tumor–bearing mice. (**E**) Quantitation of tumor ROIs from PET in (D). (**F**) Quantitation of tumor ROIs over time for the unblocked group in (D). (**G**) An Ab1-^{225}Ac conjugate was injected into HT-1080 xenograft–bearing NSG mice at 0.6 μCi on days 0 and 7. $N = 8$ mice for phosphate-buffered saline (PBS) arm and 16 mice for treatment arm. Representative of two independent studies. Dotted lines represent the tumor volume range limits. For (C), statistics were calculated by multiple *t* tests corrected for multiple comparisons. For (E), statistics were calculated by Student's *t* test. For (G), statistics were calculated with a mixed-effects model. Corrected *P* values are shown adjacent to the comparison. %ID/g, percent of decay-corrected injected dose per gram; %ID/cc, percent of decay corrected dose per cubic centimeter of the ROI. Error bars correspond to SDs, and values represent means. Imaging studies are representative of at least four animals. Blocking, biodistribution, and efficacy studies were performed twice.

We next investigated the potential of Ab1 as an antibody-drug conjugate (ADC). We were particularly interested in whether an Ab1 ADC would be cytotoxic because there is no evidence that eSrc would be internalized through a specific pathway. To assay this, we conjugated Ab1 to monomethyl auristatin E (MMAE), a tubulin inhibitor and clinical ADC payload, using nonspecific lysine labeling. Upon treatment of cancer cell lines with an Ab1-MMAE conjugate, we observed dose-dependent, albeit modest, cytotoxicity in KRAS-transformed cells that expressed eSrc but not disease-matched KRAS wild-type cells, even when those cells expressed comparable levels of eSrc (fig. S19, A to C). We hypothesized that internalization could be possible through macropinocytosis, a pathway known to be up-regulated in KRAS-transformed cell lines (*57*). To assay this, we treated KRAS-transformed KP4 cells with the macropinocytosis inhibitor ethylisopropylamiloride hydrochloride (*57*) and found that this abrogated the cytotoxic effects of Ab1-MMAE (fig. S19D). We further tested the efficacy Ab1-MMAE against KP4 xenograft tumors implanted in NSG mice in vivo; however, we did not see substantial efficacy, likely owing to the comparatively low receptor count of eSrc, limited internalization, and correspondingly modest in vitro potency compared with clinical ADCs (fig. S19, E and F).

We also evaluated eSrc as a target for T cell engagers (TCEs) by producing a bispecific antibody bearing two different CD3 binding arms that engage T cells and a Src binding arm containing the Fab from Ab1. We observed TCE-dependent cytotoxicity (specific lysis) across multiple cell lines using PBMCs from multiple healthy donors in a manner that correlated with eSrc abundance; the highest degree of specific lysis was observed for eSrc-expressing HCC1569 cells, and no specific lysis was observed for eSrc-negative HL-60 cells (Fig. 6A; fig. S20, A to E; and movies S1 to S4). Similar results were obtained in bladder cancer cell lines ranging from high to low eSrc expression (fig. S20F).

We observed potent efficacy using three TCE formats: knob-in-hole bispecific antibody, Fab-scFv fusion, and an immunoglobulin G (IgG)–scFv fusion. Monovalent controls containing an Fc bearing only a single anti-Src, or anti-CD3 arm, or a TCE bearing a Src nonbinding point mutant were far less effective at mediating specific lysis against HCC1569 cells (Fig. 6B). We hypothesized that reductions in eSrc levels with pharmacological reagents could desensitize cells to treatment with our Src-targeting TCEs. Indeed, high-eSrc HCC1569 cells could be protected from the TCE by pretreatment with IMP-1088, which reduced eSrc expression (Fig. 6C). Furthermore, low-eSrc PaTu8902 cells could be sensitized to TCE-mediated cytotoxicity by pretreatment with trametinib, which induced Src translocation through ALE (Fig. 6D). We tested the efficacy of our TCE in a humanized mouse model in which human CD8 T cells were engrafted into UM-UC-1 xenograft-bearing NSG mice 1 day before TCE administration (Fig. 6, E and F). We observed that the anti-Src TCE inhibited tumor growth in vivo. We additionally observed tumor growth inhibition or regression in two additional cancer cell line xenografts in humanized mice (fig. S21). Thus, eSrc is a therapeutically relevant target for antibody-based modalities, specifically TCEs and RLTs. These modalities elicited tumor regression in vivo and were safe to administer, even though Ab1 is cross-reactive for murine (e)Src.

Discussion

Src kinase is essential for cell viability and was one of the first identified regulators of cell proliferative activity. Src coordinates nearly all the hallmarks of cancer, particularly proliferative signaling and dysregulating cellular metabolism (*27*, *58*). Understanding the biological consequences and mechanisms of Src activity is important for developing therapeutics that may benefit a wide variety of cancer patients. Even though direct oncogenic mutation of Src is comparatively rare (1%) (*58*), Src hyperactivity and protein-level amplification are present in a wide variety of cancers. In this work, we describe a distinctive consequence of Src overexpression coupled with ALE that leads to the display of eSrc on the outer cell membrane of cancer cell lines, cell line xenografts, syngeneic tumor models, colorectal cancer and bladder cancer patient–derived xenografts, and primary tumor samples from bladder cancer patients.

Our data suggest that eSrc can help drive proliferation in cancer cells. This observation is congruent with prior work by Backe *et al.* (*6*), in which the authors observed that anti-Src function-blocking antibodies

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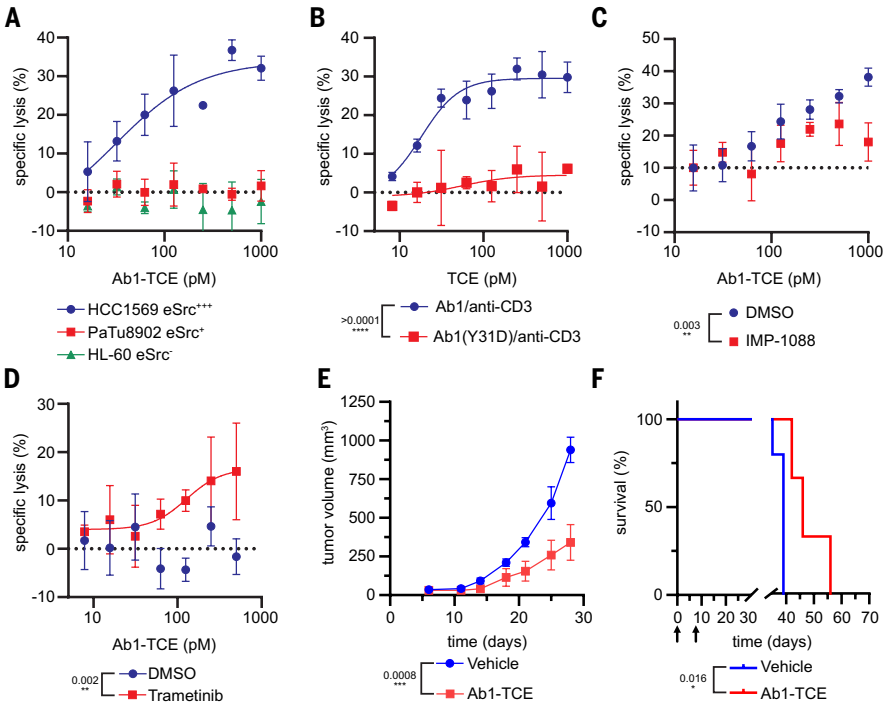

Fig. 6. eSrc is a suitable target for TCEs. (**A**) A bispecific antibody consisting of Ab1 Fab on one arm and an anti-CD3 scFv derived from clone SP34 (TCE) was assayed for efficacy at engaging T cells from PBMCs against target cells. Statistics were determined by two-way ANOVA corrected for multiple comparisons. The TCE was added to PBMCs in coculture with target cells in a standard 10:1 (E/T) ratio. Target cells were prelabeled with a live cell dye by culturing for 72 hours in the presence of a DNA-binding dye, Sytox DeepRed, impermeant to the membranes of live cell but permeable to dead cells. Images were acquired every 3 hours on an Incucyte SX5. Specific lysis is defined as lysis of the target cell in the presence of TCE minus the mean of the cells treated with PBMCs only. All data are representative of at least two independent experiments with different individual PBMC donors. (**B**) Efficacy of TCE consisting of Ab1 or a loss of Src binding point mutant corresponding to Ab1 heavy chain Y31D on HCC1569 cells. Statistics were calculated by two-way ANOVA. (**C**) HCC1569 cells were pretreated with IMP-1088 (100 nM) for 48 hours followed by drug washout before coculture with PBMC and TCE addition. Statistics were calculated by two-way ANOVA. (**D**) PaTu8902 cells were pretreated with the lysosomal exocytosis inducer, trametinib (50 nM), for 48 hours followed by drug washout before coculture with PBMC and TCE addition. Statistics were calculated by two-way ANOVA. (**E** and **F**) TCE in vivo efficacy in humanized NSG mice. UM-UC-1 tumors were implanted in NSG mice and grown to a size of 50 to 100 mm^3, at which time CD8+ T cells were intravenously injected followed by TCE administration (2.5 mg/kg, intravenously) or saline the next day and 7 days later. Five mice per arm were used. (E) Tumor volumes were measured periodically. Statistics were determined by two-way ANOVA. (F) Survival curves for the mice. Statistics were determined by Gehan-Breslow-Wilcoxon test. Error bars represent SDs, and values represent means. In vitro assays are representative of at least three independent experiments; in vivo studies use at least five mice per arm and are single studies.

elicited an antiproliferative effect. Our cell surface phosphotyrosine-containing protein enrichment method confidently identified numerous eSrc substrates, including ADAM17. This may help to provide a missing link between Src function and ADAM17 activity, as previously observed (*21*, *23*). We believe this phosphoproteomic workflow can be used to study the role of eSrc on other cell lines and can be expanded to other ectokinases, including those we have identified in this study, such as eHck.

Recently, Src has been described as an ectokinase (*6*, *7*), but the mechanism by which Src is secreted has remained elusive. In this work, we detail investigations into both the trafficking and localization of Src in the extracellular space. We show that Src, associated with autophagosomes, is translocated to the lysosome for vesicle fusion and then exported by lysosomal exocytosis. Others have proposed that, in the context of high autophagy in cancer, the plasma membrane can be used as a lipid reservoir for phagophore biogenesis (*39*–*41*). We propose a model where vesicles budded from the plasma membrane are elongated and form the nucleating phagophore membrane, with contents of the inner leaflet of the plasma membrane being presented on both the interior and exterior of autophagosomes after closure of the phagophore. This would explain the presence of Src on the interior of autophagosomes and, ultimately, on the cell surface. Our data suggest that eSrc exists in a permanently anchored state through its *N*-myristoyl lipid. This is consistent with the observations from previous work (*6*, *7*), in which the N-terminal domain was shown to be key for extracellular Src activity and detection in cell culture media. Notably, transmembrane proteins seem to be excluded by some mechanism from phagophore biogenesis (*59*). Although we observed the inversion of Src in MCF10A cells, an immortal but nominally noncancerous cell line, we saw reduced levels in isogenic oncogene-transformed systems with similar total protein expression and absent inversion, despite protein expression in primary HUVECs, which suggests that the incidence of ALE may depend on additional oncogenic or regulatory factors. Furthermore, we demonstrate that eSrc is not the only protein inverted by this process; we identified 34 other *N*-myristoylated inner leaflet proteins on the surface of other cancer cell lines, notably eHck. Our analysis did not extend to other forms of lipidation, though inversion of other membrane-anchored proteins may be possible, but is speculative at this point.

Finally, we have demonstrated that noncanonical translocation of Src to the cell surface provides a distinct opportunity for antibody-based targeting of this inverted protein. Owing to metabolic dysregulation in cancer, both Src protein levels and autophagic flux are up-regulated across a wide variety of cancers (*58*, *60*). We hypothesize that eSrc should be a highly conserved and highly tumor-associated antigen. Furthermore, based on the data herein, we estimate that the amount of eSrc on the surface of cancer cells can exceed the amount of total Src protein in healthy tissue by at least two orders of magnitude (*53*). These studies support eSrc as a highly attractive cancer-associated antigen for antibody-based therapies. We observed exceptional tumor localization with an immuno-PET in vivo imaging study and therapeutic efficacy using anti-Src antibody-based therapies. eSrc thus represents a clinically relevant cancer-associated antigen, which provides a prototypical member of a class of *N*-myristoylated inverted proteins that are noncanonically translocated owing to metabolic dysregulation in cancer with potential as therapeutically relevant cancer targets.

Materials and methods are available as supplementary materials.

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SUPPLEMENTARY MATERIALS

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Autophagolysosomal exocytosis inverts Src kinase onto the cell surface in cancer

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Editor's summary

Discovering cell surface proteins that are specific to cancer cells and not normal cells can create opportunities for safer delivery of toxic drugs to tumors. Delaveris *et al*. identified a family of proteins that are delivered to the cell surface in cancer but not in healthy tissues (see the Perspective by Pfannenstein and Meyer). One such protein is the cancer-promoting enzyme Src. The researchers developed antibodies that target the surface form of Src and showed that these anti-Src antibodies could specifically deliver toxic radioactive isotopes or recruit killer T cells to tumors in vivo. This work highlights the potential to increase the types of cell surface targets available for cancer therapies. —Stella M. Hurtley

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